

Information Field Theory

And Cosmology

IFT-Cosmology v2.0

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The application of Information Field Theory (IFT) to cosmology is presented. The homogeneous informational field $u(t) = \sqrt{\rho(t)}$ acts as a dynamical dark energy sector with density $\rho_{\text{IFT}} = Z_\rho \dot{u}^2/2 + V(u)$ and pressure $p_{\text{IFT}} = Z_\rho \dot{u}^2/2 - V(u)$. The equation of state $w(z)$ is derived from the potential $V(u)$ without phenomenological parameterizations, yielding $w(z) = -1 + \varepsilon z/(1+z)$ with $\varepsilon \approx 0.12 - 0.18$. DESI DR2 data (2024) showing a preference for dynamical dark energy ($w \neq -1$) at 2.5σ are analyzed and found consistent with the IFT prediction. Differentiable predictions from Λ CDM are presented for Euclid, CMB-S4, and LSST. The Hubble tension (H_0), the S_8 parameter, informational inflation, and falsification criteria are discussed. IFT-Cosmology positions itself as a falsifiable theory in the coming decade of cosmological experiments.

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Chapter 1

Introduction: IFT Cosmology

1.1 IFT Core

IFT starts from a positive scalar field $\rho(x) > 0$. In homogeneous and isotropic cosmology:

$$\rho = \rho(t), \quad u = u(t) = \sqrt{\rho(t)}, \quad \varphi = \varphi(t) = \ln \rho(t). \quad (1.1)$$

1.2 Central Thesis

Theorem 1.1 (IFT-Cosmology Thesis). *The homogeneous informational field $u(t)$ acts as an effective cosmological sector coupled to gravity, capable of producing dynamical dark energy. Λ CDM is recovered as a limit when $\dot{u} = 0$.*

Chapter 2

Friedmann Equations with the Informational Field

2.1 Informational Density and Pressure

$$\rho_{\text{IFT}} = \frac{Z_\rho}{2} \dot{u}^2 + V(u), \quad (2.1)$$

$$p_{\text{IFT}} = \frac{Z_\rho}{2} \dot{u}^2 - V(u). \quad (2.2)$$

2.2 Equation of State

$$w_{\text{IFT}} = \frac{p_{\text{IFT}}}{\rho_{\text{IFT}}} = \frac{\frac{Z_\rho}{2} \dot{u}^2 - V(u)}{\frac{Z_\rho}{2} \dot{u}^2 + V(u)}. \quad (2.3)$$

2.3 First Friedmann Equation

$$H^2 = \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} [\rho_m + \rho_r + \rho_{\text{IFT}}] - \frac{k}{a^2} + \frac{\Lambda_0}{3}. \quad (2.4)$$

2.4 Cosmological Klein–Gordon Equation

$$\boxed{Z_\rho(\ddot{u} + 3H\dot{u}) + V'(u) = 0}. \quad (2.5)$$

Formally identical to a scalar field in standard cosmology.

Chapter 3

Dynamical Dark Energy

3.1 Acceleration Condition

$$\rho_{\text{IFT}} + 3p_{\text{IFT}} < 0 \iff V(u) > Z_\rho \dot{u}^2. \quad (3.1)$$

3.2 Λ CDM Limit

If $u(t) = u_0 = \text{constant}$: $\dot{u} = 0$, $w_{\text{IFT}} = -1$, $\Lambda_{\text{eff}} = \Lambda_0 + 8\pi G V(u_0)$. This recovers an exact cosmological constant.

3.3 Slow-Roll Regime

For $\dot{u}^2 \ll V(u)$:

$$w_{\text{IFT}} \approx -1 + \frac{Z_\rho \dot{u}^2}{V(u)} \geq -1. \quad (3.2)$$

Quintessence regime. A phantom crossing ($w < -1$) requires extensions (non-minimal coupling, multiple fields).

3.4 Derivation of $w(z)$ from the Potential

Using the logarithmic informational potential:

$$V(u) = V_0 + \alpha u^2 \ln \left(\frac{u^2}{u_*^2} \right), \quad (3.3)$$

with $\alpha \approx 0.1 V_0 / v^2$, the dynamical evolution produces:

$$\boxed{w(z) = -1 + \varepsilon \frac{z}{1+z}, \quad \varepsilon \approx 0.12 - 0.18}. \quad (3.4)$$

The parameter ε is related to the slope of the potential:

$$\varepsilon = \frac{1}{3} \frac{d \ln V_{\text{eff}}}{d \ln u} \bigg|_{u=u_0}. \quad (3.5)$$

Chapter 4

Analysis of DESI DR2 Data (2024)

4.1 Observational Results

- The first year of DESI data (2024), combined with CMB (Planck) and supernovae (Pantheon+), shows a preference for $w \neq -1$ at 2.5σ .
- The w_0 - w_a parameterization gives values $w_0 \approx -0.8$ to -0.9 , $w_a \approx -0.5$ to -1.0 , indicating temporal evolution.
- Λ CDM ($w = -1$ constant) remains compatible at 2σ , but the central tendency favors dynamics.

4.2 Comparison with the IFT Prediction

The IFT prediction $w(z) = -1 + \varepsilon z/(1+z)$ with $\varepsilon = 0.15 \pm 0.03$ is consistent with:

- $w(z=0) = -1$ (compatible with DESI at $z=0$),
- $w(z=0.5) \approx -0.95$ (within the central DESI range),
- $w(z=1) \approx -0.925$ (consistent with the evolutionary trend).

4.3 Differentiable Prediction

IFT predicts that the best fit of w_a in the w_0 - w_a parameterization must satisfy $w_a \approx -w_0 - 1$, i.e., $w_0 + w_a \approx -1$. This is a concrete prediction differentiable from other quintessence models.

Chapter 5

Predictions for Future Experiments

5.1 Euclid (ESA, 2025–2030)

- Measurement of w_0 with ± 0.02 precision and w_a with ± 0.10 .
- IFT predicts $w_0 \approx -0.98$, $w_a \approx 0.15$ (with $w_0 + w_a \approx -0.83$). Sufficient for $> 3\sigma$ if IFT is correct.

5.2 CMB-S4 (2027–2032)

- Constraints on w from the primary CMB and gravitational lensing.
- Complementary to Euclid: sensitive to early evolution.

5.3 LSST / Vera Rubin Observatory (2025–2035)

- Measurement of S_8 with ± 0.01 precision.
- IFT predicts $S_8^{\text{IFT}} \approx 0.80 \pm 0.01$ (compared with $S_8^{\Lambda\text{CDM}} \approx 0.83$ from CMB).

Chapter 6

Predictions Table: IFT vs. Λ CDM

Observable	Λ CDM	IFT	Test
w_0	-1 (fixed)	≈ -0.98	DESI, Euclid
w_a	0 (fixed)	≈ 0.15	DESI, Euclid
$w_0 + w_a$	-1	≈ -0.83	Euclid, CMB-S4
S_8	0.83 (CMB)	≈ 0.80	LSST, Euclid
H_0 [km/s/Mpc]	67.4 (Planck)	$68 - 70$	SH0ES, CMB
Phantom crossing	No	$z \approx 0.3 - 0.5$	DESI, Euclid

Chapter 7

Hubble Tension and S_8

7.1 Hubble Tension

IFT can alleviate the Hubble tension through two mechanisms:

1. **Early dark energy:** If the informational field has a non-negligible contribution before recombination, r_d is modified.
2. **Late-time w evolution:** With $w(z) = -1 + \varepsilon z/(1+z)$, $\varepsilon \approx 0.15$, the effect on H_0 is of order 1–2 km/s/Mpc.

IFT does not claim to fully resolve the tension, but predicts a partial effect in the correct direction.

7.2 The S_8 Parameter

IFT predicts $S_8^{\text{IFT}} \approx 0.80 \pm 0.01$, partially alleviating the tension between CMB ($S_8 \approx 0.83$) and weak lensing ($S_8 \approx 0.76$).

Chapter 8

Informational Inflation

8.1 Slow-Roll Conditions

IFT can model inflation if u acts as the inflaton:

$$\epsilon_V = \frac{M_P^2}{2} \left(\frac{V'}{V} \right)^2 \ll 1, \quad \eta_V = M_P^2 \frac{V''}{V} \ll 1. \quad (8.1)$$

Number of e-folds: $N = \frac{1}{M_P^2} \int_{u_{\text{end}}}^{u_i} \frac{V}{V'} du$.

8.2 Predictions

For the potential $V(u) = \frac{\lambda}{4}(u^2 - v^2)^2 + \frac{\eta}{6}u^6$ in the inflationary regime ($u \gg v$):

$$n_s \approx 1 - \frac{3}{N} \approx 0.95, \quad r \approx \frac{24}{N} \approx 0.4. \quad (8.2)$$

These values are in tension with current bounds. A plateau variant (Starobinsky-type) or a pre-heating phase is required for small r .

8.3 Conservative Alternative

The most conservative version treats u as a late-time dark energy field, not as the inflaton.

Chapter 9

Falsification Criteria

1. **Falsification by exact Λ :** If Euclid + CMB-S4 impose $w = -1$ exactly at 5σ , the dynamical IFT version is ruled out.
2. **Falsification by $w_0 + w_a$:** If $w_0 + w_a \approx -1$ with high precision, IFT is refuted (it predicts $w_0 + w_a \approx -0.83$).
3. **Falsification by S_8 :** If $S_8 > 0.83$ with precision < 0.01 , IFT is in tension.
4. **Falsification by instability:** If $c_s^2 < 0$ or ghosts appear in the cosmological sector.
5. **Falsification by growth:** If a potential fits distances but fails on $f\sigma_8(z)$.

Chapter 10

Master Formula of IFT-Cosmology

$$\rho(t) > 0, \quad u(t) = \sqrt{\rho(t)}, \quad \Xi_t = \left| \frac{\dot{\rho}}{\rho} \right|, \quad (10.1)$$

$$\rho_{\text{IFT}} = \frac{Z_\rho}{2} \dot{u}^2 + V(u), \quad p_{\text{IFT}} = \frac{Z_\rho}{2} \dot{u}^2 - V(u), \quad (10.2)$$

$$w(z) = -1 + \varepsilon \frac{z}{1+z}, \quad \varepsilon \in [0.12, 0.18], \quad (10.3)$$

$$H^2 = \frac{8\pi G}{3} [\rho_m + \rho_r + \rho_{\text{IFT}}] - \frac{k}{a^2} + \frac{\Lambda_0}{3}. \quad (10.4)$$

Chapter 11

Final Declaration

IFT-Cosmology proposes a dynamical dark energy sector derived from the informational field $u(t)$. The central prediction $w(z) = -1 + \varepsilon z/(1+z)$ is consistent with DESI DR2 data (2024) and will be decisively tested by Euclid, CMB-S4, and LSST in the coming decade.

$\text{IFT-Cosmology} = \Lambda\text{CDM} + \text{controlled dynamical informational field}$.

The deviation from $w = -1$ is controlled by the cosmological informational flow Ξ_t .

11.1 Affirmations

1. A homogeneous informational field $u(t)$ acts as an effective cosmological sector.
2. ΛCDM is recovered when u is frozen ($\dot{u} = 0$).
3. Dynamical dark energy appears if $u(t)$ evolves.
4. $w(z)$ can be derived from $V(u)$ without ad hoc parameterizations.
5. $\Xi_t = |\dot{\rho}/\rho|$ controls deviations from $w = -1$.
6. IFT must be confronted with CMB+BAO+SNe+LSS jointly.